

TradeAux

Global Commodity Auction Platform

Business Assessment Report

Prepared: May 2026 | Stage: Pre-Revenue / MVP Build

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1. Executive Summary

Overview

TradeAux is an early-stage B2B commodity auction platform that eliminates intermediaries by directly connecting agricultural co-operatives, food manufacturers/importers, and freight carriers in a three-sided marketplace. Its primary innovation is the **Aggregate Sale** mechanism — splitting a single seller listing across multiple buyers — and a lean, flat-fee revenue model (\$25 per winning bid) that aligns costs with transaction success.

Core Thesis: TradeAux captures value at the intersection of three underserved markets — surplus commodity supply, global buyer demand, and empty carrier capacity — using a technology-light fee structure that scales with volume, not complexity.

Metric	Detail
Business Stage	Pre-revenue, MVP frontend complete, Firebase integration in progress
Primary Market	Coconut co-ops (30 target co-ops), expanding to broader agri-commodities
Revenue Model	\$25 flat fee per winning bid; no subscription, no listing fees
Break-even Target	600 winning bids/month = \$15,000 MRR
12-Month ARR Target	\$180,000+
Key Differentiator	Aggregate Sale: one listing generates 20-30 x \$25 fees vs. single sale
Technology Stack	Firebase (Firestore, Auth, Cloud Functions, Hosting), Vanilla JS frontend
Critical Risk	Three-sided marketplace cold start — chicken-and-egg bootstrapping challenge
Overall Assessment	Strong concept, viable model; execution risk is high without funding & sales

2. Business Model Analysis

The Three-Sided Marketplace

TradeAux operates a three-sided network — sellers, buyers, and carriers — each solving a distinct pain point. This structure creates strong cross-side network effects: more sellers attract more buyers, more buyers attract more carriers to fill transport needs, and carrier density makes the platform more attractive to sellers needing logistics.

Participant	Pain Point Solved	Value Proposition
Sellers (Co-ops / Factories)	Sell to middlemen at suppressed prices; limited global reach; surplus goes to waste	Direct global buyer access; competitive price discovery; monetise off-grade inventory
Buyers (Importers / Manufacturers)	Pay broker premiums; opaque pricing; limited supplier diversity	Competitive bidding drives prices down; direct supplier relationships; transparent grades
Carriers (Shipping / Freight)	Run empty backhauls; costly excess capacity; fragmented cargo sourcing	Fill dead-leg capacity with matched cargo; incremental revenue on sunk costs

Fee Structure Mechanics

The fee architecture is intentionally frictionless. TradeAux collects only the buyer's \$25 flat fee in cash. Seller and carrier discounts are peer-to-peer, reducing TradeAux's payment-processing complexity at early stage. This design lowers the barrier to platform adoption but caps per-transaction revenue.

Role	Trigger	Fee / Discount	Note
Buyer	Wins a bid	\$25 cash fee to TradeAux	Low friction — fixed cost
Seller	Wins a buyer brief	\$50 discount off invoice	No cash to TradeAux
Carrier	Wins transport auction	\$25 discount off freight invoice	No cash to TradeAux

Aggregate Sale Advantage: One seller listing of 500 MT can be split across 20-30 buyers (partial quantities). Each acceptance triggers a \$25 fee, turning a single listing into \$500-\$750 of platform revenue. This is the core economic multiplier of TradeAux.

The Seller Bot — Optimization Engine

Rather than a simple 'highest bid wins' approach, the Seller Bot collects all bids on a listing and runs a value-optimization algorithm to accept the combination of partial bids that maximises total invoice value. Each accepted partial bid generates a fee. This is technically sophisticated but operationally unproven — validation will require real bid data.

3. Revenue Model & Financial Projections

Revenue Mechanics

Metric	Value
Revenue per winning bid	\$25
Break-even (monthly)	600 bids / month
Break-even (annual)	7,200 bids / year
Target monthly revenue	\$15,000
Target ARR (run rate)	\$180,000+
Revenue per listing (30 buyers)	\$750
Revenue per co-op (est.)	\$500 – \$1,500 / month
Co-ops needed for \$15k MRR	~10–30 active co-ops

Growth Scenarios

Scenario	Co-ops	Bids/Listing	Listings/Co-op	Total Bids/Mo	MRR
Conservative	5 co-ops active	10 bids/listing avg	1 listing/co-op/mo	50 bids/mo	\$1,250/mo
Moderate	15 co-ops active	15 bids/listing avg	3 listings/co-op/mo	675 bids/mo	\$16,875/mo
Optimistic	30 co-ops active	20 bids/listing avg	5 listings/co-op/mo	3,000 bids/mo	\$75,000/mo

Revenue Model Assessment

- **Strength:** Zero marginal cost per additional bid — once infrastructure is built, each \$25 fee is near-pure margin.
- **Strength:** Aggregate Sale multiplier is a genuine competitive moat — not easily replicated by simple auction software.
- **Weakness:** \$25/bid is very low. A 600-bid/month target requires significant transaction volume to sustain operations.
- **Weakness:** Reliance on buyer-side cash fees only; seller and carrier fees are discounts, not collected revenue. Future monetisation of these sides represents significant upside but requires payment infrastructure.
- **Opportunity:** Premium tiers (featured listings, verified co-op badges, analytics) could 3-5x ARPU without volume growth.

- **Risk:** Race to zero — buyers may demand lower fees as alternatives emerge. Fee lock-in strategy needed early.

4. SWOT Analysis

STRENGTHS	WEAKNESSES
• Aggregate Sale mechanism is a genuine IP advantage • Flat fee model is simple and friction-free • Three-sided network creates compounding value • Solves real, documented pain for all three parties • Geographic exclusion feature prevents channel conflict • Firebase stack is lean, scalable, and low-cost	• Three-sided cold-start problem is extremely hard • \$25 fee is low — needs high volume for meaningful revenue • No payment rails for seller/carrier discounts yet • Seller Bot (optimization engine) is unproven • Single developer/founder — execution bandwidth risk • No traction or proof of concept with real users yet
OPPORTUNITIES	THREATS
• \$3T+ global commodity trade largely broker-mediated • Coconut co-ops underserved — fast initial traction possible • Expand beyond coconuts to other agri-commodities • Carrier side is virtually untapped (empty backhauls) • Premium SaaS tiers (analytics, verified badges, APIs) • ESG/fair-trade angle appeals to impact investors	• Established B2B marketplaces (Alibaba, Agri-Marketplace) pivoting to auctions • Regulatory complexity in cross-border commodity trade • Co-op members may resist technology adoption • Currency and payment risk across markets • Trust deficit — new platform, no track record • Sellers reverting to direct deals off-platform ('leakage')

5. Market Assessment

Target Market: Coconut Co-ops

The initial focus on large coconut co-ops is strategically sound. The global coconut products market exceeds \$15B annually, with key production concentrated in the Philippines, Indonesia, Sri Lanka, and India — all regions with highly fragmented supply chains dominated by brokers. Co-ops represent aggregated supply, making them ideal first sellers because a single relationship unlocks volume.

Metric	Estimate
Global Coconut Products Market	~\$15B+ annually
Key Producing Regions	Philippines, Indonesia, Sri Lanka, India
Target Co-ops (Phase 1)	30 large coconut co-ops
Estimated Revenue / Co-op	\$500 – \$1,500 / month
Time to \$15k MRR	3 – 6 months (with 10-30 active co-ops)
Total Addressable Market (TAM)	\$500M+ (global agri-auction / B2B platform segment)
Serviceable Addressable Market	\$50M (coconut + select tropical commodities, 5-yr)
Beachhead Market	\$1.8M ARR from 30 co-ops at target volume

Market Entry Assessment

- **Favourable:** Co-ops are frustrated with broker dependency — the value proposition is immediately understood.
- **Favourable:** Export-grade surplus and off-grade inventory are genuine pain points with no good current solution.
- **Challenge:** Co-op decision cycles are slow — committee-driven approvals may delay onboarding.
- **Challenge:** Technology literacy varies significantly across co-op membership; platform must be very simple.
- **Favourable:** Geographic exclusion feature directly addresses channel conflict fears — a common blocker for co-ops joining marketplaces.
- **Expansion Path:** Coffee, cacao, palm oil, spices, and dried fruits share identical structural pain points and can be layered in post-coconut.

6. Technical Architecture Review

Stack Assessment

Layer	Technology	Status	Notes
Frontend	Vanilla JS / HTML5 / CSS3	Adequate for MVP	No framework overhead. Future scale may require React/Vue for component reuse.
Hosting	Firebase Hosting	Good choice	Global CDN, zero-config SSL, tight Firestore integration.
Database	Firestore (NoSQL)	Good choice	Real-time listeners ideal for live auction bid updates.
Auth	Firebase Auth	Good choice	Role-based redirect post-login is correct pattern.
Backend Logic	Cloud Functions (Node.js)	Adequate	Correct for bid validation, but cold-start latency may affect bid-placement UX.
File Storage	Firebase Storage	Good choice	For product images — straightforward and integrated.
Payments	Not started	Critical gap	Stripe or Wise needed. No revenue collection possible without this.
Email / Notify	Not started	High priority	Bid confirmations, auction alerts — essential for user engagement.

Completion Status

Item	Status	Notes
Business model & fee structure	Complete	Foundation solid
Frontend pages (HTML/CSS/JS)	Complete	All key pages exist
Firebase project setup	Complete	tradeaux-c72c7
Firebase Auth (sign-in flow)	Complete	Role-based redirect working
Firestore data seeding	In Progress	Test data needed
placeBid Cloud Function	In Progress	Core bidding logic pending
Auction floors — Firestore connect	In Progress	3 floors need wiring
Bid acceptance (seller side)	Not Started	Critical path item
Wallet system	Not Started	Needed for fee tracking
Firestore security rules	Not Started	Security risk until done
Firebase Hosting deployment	Not Started	Platform not yet live
Payment processing (Stripe/Wise)	Not Started	Revenue blocker

Item	Status	Notes
Email notifications	Not Started	UX critical
Seller Bot automation	Not Started	Core differentiator
Photo upload (Firebase Storage)	Not Started	Trust-building feature

7. Competitive Landscape

Key Competitors & Alternatives

Competitor	Positioning	TradeAux Differentiation	Threat Level
Alibaba / Made-in-China	B2B commodity listings	No live auction; no carrier side; broker-mediated; high listing fees	Indirect
AgriDigital (AU)	Grain supply chain platform	Commodity-specific (grain); no multi-buyer aggregate sale; AU-focused	Indirect
Tridge	Global agri trade intelligence + sourcing	Data-heavy, broker-assisted; no auction mechanism; subscription model	Indirect
Cargill Direct / ADM	Direct procurement platforms	Internal tools of large buyers; not open marketplace; excludes small sellers	Indirect
Local commodity brokers	Traditional intermediaries	High margin extraction; no tech; entrenched relationships — hardest to displace	Direct
Custom buyer portals	Large importers with own sourcing portals	Closed ecosystems; don't solve seller/carrier fragmentation	Partial

TradeAux's core moat is NOT the technology — it is the Aggregate Sale mechanism combined with the three-sided network. No current competitor combines live auction + partial-fill optimization + carrier integration in one platform. This is the defensible wedge.

8. Risk Register

Risk	Severity	Description	Mitigation
Cold Start Problem	Critical	Three-sided marketplace requires simultaneous seller, buyer & carrier onboarding	Seed platform with 5 known co-ops + pre-recruited buyer pool before public launch
Platform Leakage	High	Sellers and buyers connect directly off-platform after first transaction	Build trust features (verified reviews, escrow), keep fees below broker cost
Payment Infrastructure	High	No Stripe/Wise integration — zero revenue collection possible	Prioritise Stripe Connect setup before any marketing spend
Firestore Security	High	No security rules set — data is exposed and writable by anyone	Implement security rules before any user onboarding
Regulatory / Customs	Medium	Cross-border commodity trade involves phytosanitary certs, import duties, CITES	Partner with licensed freight forwarders; legal disclaimer; country-specific guides
Single-Founder Execution	Medium	One developer building product + selling + ops simultaneously	Hire or partner for BD/sales; focus technical work on critical path only
Seller Bot Reliability	Medium	Optimization algorithm could produce suboptimal or legally questionable splits	Manual override for sellers; thorough testing with synthetic bid data first
Currency & FX Risk	Low-Medium	Multi-currency transactions with exchange rate exposure	USD-denominated listings; integrate FX layer (Wise) before multi-currency
Co-op Technology Adoption	Low-Medium	Committee approval slow; member literacy varies	White-glove onboarding; WhatsApp-based notifications; local language support

9. Go-to-Market Strategy

Phase 1 — Beachhead (Months 1-3): Coconut Co-ops

- Identify and personally onboard **3-5 large coconut co-ops** with pre-existing export relationships.
- Simultaneously recruit **15-20 verified buyers** (importers, food manufacturers) who already source coconut products.
- Seed the auction floor with real listings from co-ops before public launch — no cold floor.
- Deploy a WhatsApp Business number for co-op notifications (bid received, auction closing soon).
- Offer **zero platform fee for first 30 days** to eliminate adoption friction.

Phase 2 — Growth (Months 3-6): Transport + Scale

- Activate **carrier onboarding** — target freight forwarders on established coconut trade lanes (Colombo, Manila, Jakarta).
- Introduce **Aggregate Sale listings** with Seller Bot once first 3 co-ops are live.
- Scale buyer pool to **50+ verified buyers** across 10+ import markets.
- Deploy email + SMS notifications to drive bid activity.
- Target: 600+ bids/month by Month 6.

Phase 3 — Expansion (Months 6-12): Multi-Commodity

- Open platform to **coffee, cacao, palm oil, spices** — same model, new commodity categories.
- Launch **premium seller tiers**: verified co-op badge, priority listing placement, export analytics dashboard.
- Explore **impact investor pitch**: ESG narrative (fair-trade, smallholder empowerment) + proven traction.
- Consider **white-label version** for commodity associations and trade bodies.

10. Scorecard & Overall Rating

The following scores reflect a qualitative assessment across key business dimensions for an early-stage marketplace platform at MVP stage.

Dimension	Score /10	Commentary
Problem / Solution Fit	8.5 [=====--]	Genuine pain points; clear value for all three sides
Business Model Viability	7.0 [=====--]	Sound, but \$25/bid is thin — needs volume or premium tiers
Market Opportunity	8.0 [=====--]	Large TAM; coconut beachhead is realistic
Competitive Differentiation	7.5 [=====--]	Aggregate Sale is unique; moat needs to be built quickly
Technical Architecture	6.5 [=====--]	Firebase stack is good; 60% of critical features not yet built
Execution Readiness	5.0 [=====--]	Single developer; no live users; no payments yet
Go-to-Market Clarity	6.0 [=====--]	Beachhead defined but no active sales pipeline confirmed
Financial Projections	6.5 [=====--]	Realistic moderate scenario; conservative scenario underwhelming
Risk Management	5.5 [=====--]	Cold start & leakage risks are existential without mitigation
Overall	6.6 [=====--]	Strong concept, high potential — execution is the gating factor

Overall Rating: 6.6 / 10 — PROMISING with HIGH EXECUTION RISK. The business model is differentiated and the market is real. The platform will succeed or fail based on the quality of sales execution and the speed at which payment infrastructure and the Seller Bot are completed.

11. Strategic Recommendations & Roadmap

Immediate Priorities (Next 30 Days)

#	Task	Priority	Rationale
1	Fix Firestore security rules	CRITICAL	Platform is insecure — do this before any user onboarding
2	Deploy placeBid Cloud Function	CRITICAL	Core auction mechanic — nothing works without this
3	Connect auction floors to Firestore	CRITICAL	End-to-end bid flow must be testable
4	Integrate Stripe Connect for \$25 fee	HIGH	Revenue collection — no business without this
5	Seed 10 real seller_briefs in Firestore	HIGH	Need real data to test and demo the platform
6	Deploy to Firebase Hosting	HIGH	Get a live URL to show co-ops and buyers

Short-Term Priorities (Days 30-90)

#	Task	Priority	Rationale
7	Onboard first 3 co-ops (pilot)	HIGH	Proof of concept; generate first real bids
8	Recruit 20 verified buyers	HIGH	Supply without demand = no bids
9	Email notification system	HIGH	Bid alerts drive engagement and return visits
10	Seller bid-acceptance flow	HIGH	Sellers need to accept/reject bids to close transactions
11	Wallet system (Firestore)	MEDIUM	Track fees collected; build toward financial reporting
12	WhatsApp Business integration	MEDIUM	Meets co-ops where they are — high ROI for adoption

Medium-Term Strategic Recommendations

- **Raise a Pre-Seed Round:** \$150k-\$300k to fund 6 months of full-time development + BD. The ESG / fair-trade angle is compelling to impact-focused angels and accelerators (e.g. Antler, Techstars Farm-to-Fork).
- **Hire a Business Development Lead:** The technical founder should not be the sole salesperson. A BD hire with co-op industry relationships accelerates onboarding by 3-5x.
- **Build the Seller Bot incrementally:** Start with simple highest-bid-wins logic, then layer in optimization. Don't let perfect be the enemy of shipped.
- **Lock in early co-op relationships with exclusivity:** Offer founding co-ops 12 months of reduced fees in exchange for exclusivity. This builds defensible supply-side density.

- **Pursue commodity association partnerships:** Philippine Coconut Authority, Indonesian Coconut Association — one partnership can unlock dozens of co-ops simultaneously.
- **Document everything for a Series A narrative:** Track bids, transaction volume, co-op earnings uplift vs. broker prices. This data is the fundraising story.

TradeAux has the ingredients of a fundable, scalable commodity marketplace. The model is differentiated, the market is real, and the technology path is clear. The single most important next step is not writing more code — it is getting the first real buyer and the first real co-op into the same auction room.